Mode 1 — Square

DP16 (8×4) → DP8 (8×4) → DP8 (4×4) → add-and-square

Step 1 — DP16 (8×4)

$$\mathrm{DP16}\,(8{\times}4) = \sum_{i=0}^{15} a_i\, b_i$$

Step 2 — DP8 (8×4)

$$\mathrm{DP16}\,(8{\times}4) = \underbrace{\sum_{i=0}^{7} a_i\, b_i}_{\mathrm{DP8}\,(8\times4)} + \underbrace{\sum_{i=8}^{15} a_i\, b_i}_{\mathrm{DP8}\,(8\times4)}$$

Step 3 — DP8 (4×4)

$$\begin{aligned}\mathrm{DP16}\,(8{\times}4) &= 2^4 \underbrace{\sum_{i=0}^{7} a_{H,i}\, b_i}_{\mathrm{DP8}\,(4\times4)} + \underbrace{\sum_{i=0}^{7} a_{L,i}\, b_i}_{\mathrm{DP8}\,(4\times4)} \\ &\quad + 2^4 \underbrace{\sum_{i=8}^{15} a_{H,i}\, b_i}_{\mathrm{DP8}\,(4\times4)} + \underbrace{\sum_{i=8}^{15} a_{L,i}\, b_i}_{\mathrm{DP8}\,(4\times4)}\end{aligned}$$

Step 4 — Add-and-square (substitute each DP8 (4×4) $= \frac{1}{2}\left[\sum (x_i{+}b_i)^2 - \sum x_i^2 - \sum b_i^2\right]$)

$$\begin{aligned}\mathrm{DP16}\,(8{\times}4) &= 2^4 \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}(a_{H,i}{+}b_i)^2 - \sum_{i=0}^{7} a_{H,i}^2 - \sum_{i=0}^{7} b_i^2\Big)}_{\mathrm{DP8}\,(4\times4)} \\ &\quad + \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}(a_{L,i}{+}b_i)^2 - \sum_{i=0}^{7} a_{L,i}^2 - \sum_{i=0}^{7} b_i^2\Big)}_{\mathrm{DP8}\,(4\times4)} \\ &\quad + 2^4 \underbrace{\tfrac{1}{2}\Big(\sum_{i=8}^{15}(a_{H,i}{+}b_i)^2 - \sum_{i=8}^{15} a_{H,i}^2 - \sum_{i=8}^{15} b_i^2\Big)}_{\mathrm{DP8}\,(4\times4)} \\ &\quad + \underbrace{\tfrac{1}{2}\Big(\sum_{i=8}^{15}(a_{L,i}{+}b_i)^2 - \sum_{i=8}^{15} a_{L,i}^2 - \sum_{i=8}^{15} b_i^2\Big)}_{\mathrm{DP8}\,(4\times4)}\end{aligned}$$

Step 5 — All components

$$\boxed{\begin{aligned}\mathrm{PE} &= 2^4 \sum_{i=0}^{7}(a_{H,i}{+}b_i)^2 + \sum_{i=0}^{7}(a_{L,i}{+}b_i)^2 + 2^4 \sum_{i=8}^{15}(a_{H,i}{+}b_i)^2 + \sum_{i=8}^{15}(a_{L,i}{+}b_i)^2 \\ \alpha &= 2^4 \sum_{i=0}^{7} a_{H,i}^2 + \sum_{i=0}^{7} a_{L,i}^2 + 2^4 \sum_{i=8}^{15} a_{H,i}^2 + \sum_{i=8}^{15} a_{L,i}^2 \\ \beta &= 2^4 \sum_{i=0}^{7} b_i^2 + \sum_{i=0}^{7} b_i^2 + 2^4 \sum_{i=8}^{15} b_i^2 + \sum_{i=8}^{15} b_i^2\end{aligned}}$$